

University of Victoria
Faculty of Engineering
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**Design and Analysis of a Pipelined RTL Hardware Accelerator
for Real-Time Sobel Edge Detection**

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1.0 Background

Digital image processing relies heavily on two-dimensional (2D) convolution, an operation in which a small coefficient matrix, known as a kernel or mask, is slid across an image and combined with the underlying pixels through multiply-accumulate arithmetic. Each output pixel is the weighted sum of a local neighbourhood of input pixels, and the kernel coefficients determine the filtering behaviour, such as blurring, sharpening, or gradient estimation [1]. Because the same arithmetic pattern repeats for every pixel, 2D convolution is computationally regular but data-intensive, which makes it a natural candidate for hardware acceleration.

Sobel edge detection is a widely used application of 2D convolution. It estimates the horizontal and vertical intensity gradients of an image using two 3×3 kernels and combines them into an edge-magnitude image that highlights boundaries between regions. Sobel filtering is favoured in embedded vision because it is computationally inexpensive relative to its usefulness, provides reasonable noise immunity through its built-in smoothing term, and maps efficiently onto fixed-function arithmetic [2].

Real-time image processing imposes a hard constraint: every pixel of every frame must be processed within a fixed time budget. A general-purpose central processing unit (CPU) executes the convolution as a sequence of instructions, and its throughput is limited by instruction fetch, memory access, and cache behaviour, all of which introduce non-deterministic latency. For embedded vision systems operating at sustained frame rates, this variability can prevent the system from meeting its timing budget.

Field-programmable gate arrays (FPGAs) and register-transfer-level (RTL) architectures address this limitation by implementing the convolution directly as dedicated digital logic. Rather than executing instructions, an RTL accelerator forms a spatial datapath that processes pixels as they stream through, exploiting both pipelining and parallelism, and allowing the designer to trade silicon area for throughput in a controlled and deterministic way [3].

Four engineering quantities dominate the design of such an accelerator. Throughput measures how many output pixels are produced per unit time and determines whether real-time frame rates can be sustained. Latency measures the delay between a pixel entering and the corresponding result leaving the pipeline, and matters for closed-loop or interactive systems. Resource usage measures the look-up tables (LUTs), flip-flops (FFs), digital signal processing (DSP) blocks, and block RAM (BRAM) consumed, which constrains cost and scalability. Timing closure refers to whether the synthesised circuit can meet its target clock period; a design that cannot close timing cannot run at the intended frequency regardless of its architectural merit.

2.0 Objective

2.1 Problem

Real-time Sobel edge detection requires deterministic, sustained processing of a continuous pixel stream. A processor-based implementation may not provide the guaranteed throughput and bounded latency needed for embedded vision at standard video resolutions and frame rates, because instruction execution and memory access introduce variability. A dedicated RTL hardware accelerator is therefore required, but it is not obvious which RTL architecture best satisfies the throughput, latency, resource, and timing requirements of the application.

2.2 Purpose

The purpose of the final report is to determine the most suitable RTL hardware architecture for a real-time 3×3 Sobel edge-detection accelerator that processes a streaming grayscale image source.

2.3 Aims

The aims of the project are to:

1. Support at least 640×480 grayscale video at 30 frames per second, equivalent to a sustained pixel rate of 9.216 million pixels per second.
2. Compare a sequential and a pipelined RTL architecture using cycles per output pixel as the primary throughput metric.
3. Estimate or measure FPGA resource usage (LUTs, flip-flops, DSP blocks, and BRAM) using Xilinx Vivado synthesis [4].
4. Determine whether the selected design can meet a clock frequency sufficient for real-time processing of the target video stream.
5. Verify the RTL output against a software reference (golden) model to a defined accuracy.

2.4 Limitations

The scope of the project is bounded as follows:

- Only grayscale (single-channel) images are processed; RGB colour processing is excluded.
- Only the standard 3×3 Sobel operator is considered; other kernels and kernel sizes are excluded.
- No physical camera interface is required; inputs are stored images or test vectors.
- No full application-specific integrated circuit (ASIC) layout is performed.
- No neural-network inference or learned filtering is included.

- The implementation may be simulated using stored image and test-vector inputs rather than live video.

3.0 Potential Solutions

3.1 Sequential RTL Filter

The sequential RTL filter loads or assembles a 3×3 pixel window and then performs the Sobel arithmetic over several clock cycles using a small number of shared arithmetic resources. A finite-state machine sequences the multiply-accumulate steps for the horizontal and vertical gradients, accumulating partial sums in registers before producing one output pixel. Because relatively few operations occur in parallel, the design uses modest logic resources and is straightforward to design, debug, and verify. Its principal limitation is throughput: each output pixel requires multiple clock cycles, so the sustained pixel rate is the clock frequency divided by the cycles per pixel, which may be insufficient for real-time operation at the target resolution and frame rate.

3.2 Pipelined RTL Filter

The pipelined RTL filter uses line buffers and shift registers to continuously assemble a 3×3 pixel window directly from the incoming pixel stream. Two line buffers store the two most recent image rows so that the three vertically aligned pixels of each window column are available simultaneously; three shift registers then provide the three horizontal taps [3]. The Sobel arithmetic is partitioned into pipeline stages so that, after an initial line-buffer and pipeline-fill delay, the circuit produces one filtered output pixel per clock cycle. This architecture is more complex and consumes more registers and memory, but it directly targets the high-throughput, low-latency, streaming nature of the problem.

4.0 Initial Assessment

The two candidate architectures are compared using a weighted decision matrix. Seven criteria were selected because they map directly onto the project aims and onto the engineering quantities introduced in Section 1.0. Each criterion is assigned a weight reflecting its importance to the objective, and each candidate is scored from 1 (poor) to 5 (excellent). The weighted score for each candidate is the sum of the products of weight and score, as defined in Equation (1).

$$S = \sum (w_i \times r_i) \quad (1)$$

In Equation (1), S is the total weighted score for a candidate architecture, w_i is the weight assigned to criterion i , and r_i is the score awarded to that candidate for criterion i . The summation is taken over all seven criteria. The scores are preliminary engineering estimates used to guide solution selection and will be revisited once Vivado synthesis results are available.

The weighting prioritises throughput and latency most heavily because the central engineering problem is sustaining a deterministic real-time pixel rate. Timing feasibility is also weighted

highly, because a design that cannot close timing cannot meet the rate regardless of its theoretical throughput. Hardware resource usage carries moderate weight, since the target device has finite logic and memory. Implementation and verification complexity are weighted lower because they affect development effort rather than whether the delivered accelerator meets its requirements. Relevance to digital-design industry skills is included as a modest educational criterion.

Table 1: Weighted decision matrix for selecting the RTL Sobel architecture.

Criterion	Weight	Seq. r	Seq. w×r	Pipe. r	Pipe. w×r
Throughput	0.25	2	0.50	5	1.25
Latency	0.15	3	0.45	4	0.60
Hardware resource usage	0.15	5	0.75	3	0.45
Timing feasibility	0.15	4	0.60	4	0.60
Implementation complexity	0.10	5	0.50	2	0.20
Verification complexity	0.10	4	0.40	3	0.30
Relevance to industry skills	0.10	3	0.30	5	0.50
Total weighted score (S)	1.00		3.50		3.90

As summarised in Table 1, the pipelined RTL filter achieves the higher total weighted score, driven mainly by its superior throughput, which is the most heavily weighted criterion. The sequential design scores well on resource usage and simplicity but is penalised heavily on throughput, the criterion most directly tied to the engineering problem. On the basis of this initial assessment, the pipelined RTL filter is selected as the proposed solution, subject to confirmation by Vivado synthesis and timing results in the later milestones.

References

- [1] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. New York, NY: Pearson, 2018.
- [2] N. Kanopoulos, N. Vasanthavada, and R. L. Baker, "Design of an image edge detection filter using the Sobel operator," *IEEE J. Solid-State Circuits*, Vol. 23, No. 2, pp. 358–367, Apr. 1988.
- [3] D. G. Bailey, *Design for Embedded Image Processing on FPGAs*. Singapore: John Wiley & Sons (Asia), 2011.
- [4] *Vivado Design Suite User Guide: Synthesis*, UG901 (v2024.1), Advanced Micro Devices, Inc., San Jose, CA, 2024.